Total Number of Pages: 01

Student Name :_____

Enrollment Number:_____

Mid Term Examination- Nov 2022

Program : B. Tech
Paper Code : ECC 205
Time Duration : $1\frac{1}{2}$ hours

Semester : Third
Paper Name : Signals and Systems
Maximum Marks : 30

Note: Question No. 1 is compulsory. Attempt any two questions from the remaining questions. Only scientific calculator is allowed.

Question 1

1(a) Check whether the system described by $y(n) = x(n) + 6$ is linear or non-linear. [2]

1(b) Sketch $x(-2t+1)$ for a given signal $x(t) = 2[u(t) - u(t-3)]$. [2]

1(c) Determine whether a signal $x(t) = e^{-a|t|}$, $a > 0$ is energy or power signal or neither. [2]

1(d) Determine the Nyquist rate of a signal $x(t) = \frac{sin(4000\pi t)}{\pi t}$. [2]

1(e) Obtain the Fourier series coefficients of a discrete time signal $x(n) = 0.5 + cos(3\pi n + \pi/3)$. [2]

Question 2

2(a) Find the output of an LTI systems whose impulse response is $h(t) = e^{-2t}u(t)$ and input is $x(t) = e^{-t+1}u(t-1)$. [5]

2(b) Find the linear convolution of two sequences $x_1(n) = [2, \underset{\uparrow}{1}, -1, 3]$ and $x_2(n) = [\underset{\uparrow}{2}, 1, 5]$. [5]

Question 3

3(a) Determine a signal $x(t)$, when the following facts about $x(t)$ are given:

1. $x(t)$ is a real and odd signal.
2. $x(t)$ is periodic with period $T = 4$, and it has exponential Fourier series coefficients c_k.
3. $c_k = 0$ for $|k| > 1$.
4. $\frac{1}{4}\int_0^4 |x(t)|^2 dt = 4$.

[5]

3(b) Consider a causal LTI system with frequency response $H(\omega) = \frac{1}{3+j\omega}$. For a particular input $x(t)$, this system is observed to produce the output $y(t) = e^{-3t}u(t) - e^{-4t}u(t)$. Determine the input $x(t)$. [5]

Question 4

4(a) Find the Fourier transform of a signal $x(t) = \frac{1}{1+t^2}$. However, the Fourier transform of $g(t) = e^{-|t|}$ is given as: $G(\omega) = \frac{2}{1+\omega^2}$. [5]

4(b) The Laplace transform of a signal $x(t)$ is given as $X(s) = \frac{3s+1}{s^3+4s^2+(k-3)s}$. If $\lim_{t\to\infty} x(t) = 1$, then find the value of k ? [5]